

Comparative Analysis of Classical and Deep Learning Methods for Unsupervised Industrial Defect Detection

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Abstract— Automated visual inspection is an important challenge in the modern manufacturing with manual defect detection being slow, expensive, and inaccurate. The paper is a detailed comparison of three methods of unsupervised anomaly detection in the MVTec Anomaly Detection (MVTecAD) benchmark dataset on all 15 categories of products. We apply and compare: (1) a classical baseline using Local Binary Patterns (LBP), Gray Level Co-occurrence Matrix (GLCM), and Canny/Sobel edge responses with Principal Component Analysis (PCA) and a One-Class Support Vector Machine; (2) a Convolutional Autoencoder that detects defects by reconstruction error; and (3) PaDiM is Patch Distribution Modeling which uses ResNet18 features trained and multivariate Gaussian model using Mahalanobis distance scoring. PaDiM has the highest ROC AUC of 0.9262 as compared to the classical baseline (0.6468) and autoencoder (0.5625) by 28% and 36% respectively. Localization at the pixel-level has a mean IoU of 0.2058. Findings indicate that trained deep features can be remarkably generalized to industrial inspection with no defect-related training.

Keywords— anomaly detection, industrial inspection, MVTec AD, PaDiM, convolutional autoencoder, one-class SVM, local binary patterns, transfer learning, unsupervised learning

I. INTRODUCTION

Quality control in manufacturing is a fundamental requirement across industries ranging from automotive to pharmaceuticals. Defects such as surface cracks, contamination, and structural deformations can lead to product failures, safety hazards, and costly recalls. Traditional manual visual inspection relies on trained human inspectors examining products on production lines — a process that is inherently slow, expensive, and subject to fatigue-induced errors.

The emergence of computer vision and machine learning offers a pathway to automated, objective, and scalable defect detection. A key challenge in this domain is the scarcity of defective examples: in real manufacturing settings, defects occur rarely and unpredictably, making it infeasible to collect comprehensive labeled defect datasets for supervised classification. This motivates the unsupervised anomaly detection paradigm, where models are trained exclusively on defect-free (normal) images and anomalies are detected as deviations from the learned normal distribution.

This paper addresses the problem of unsupervised industrial defect detection using the MVTec Anomaly Detection (MVTec AD) benchmark [1], a comprehensive real-world dataset containing 15 different product categories with pixel-level ground truth defect annotations. We implement and systematically compare three methodologically distinct approaches at increasing levels of complexity: a classical feature engineering pipeline, a deep

autoencoder, and PaDiM — a state-of-the-art patch distribution modeling approach [2].

Our evaluation demonstrates that PaDiM achieves a mean ROC AUC of 0.9262 — a substantial improvement of 28% and 36% over the classical and autoencoder baselines respectively. Pixel-level defect localization yields a mean IoU of 0.2058. These results underscore the critical role of pretrained deep features in unsupervised industrial inspection.

II. BACKGROUND

A. Anomaly Detection Paradigm.

Unsupervised anomaly detection assumes that the identification of defects is a one-class classification problem. These methods do not use balanced datasets of defects, whereas supervised methods demand balanced datasets of defects, and they train a compact representation of normal data and consider anomalies as deviations. This model fits well with industrial inspection where there are very few defects, and they are varied [1].

B. Classical Approaches

Conventional computer vision approaches to texture analysis are based on Local Binary Patterns (LBP) [5] which encode local texture by comparing pixel neighborhoods, and Gray Level Co-occurrence Matrix (GLCM) features [6] which encode spatial relationships between pixel intensities. With edge detectors (Canny, Sobel) and statistical moments, these attributes give the chance to create compact image descriptors that can be inputted to an anomaly detector of classical methods like One-Class SVM [7]. These techniques are computationally efficient, but do not have the representational ability to deal with the complex, high-resolution industrial textures.

C. Autoencoder-Based Detection

Convolutional autoencoders have found extensive use in the paradigm of anomaly detection, based on the reconstruction error paradigm [3]. The model is trained to encode and decode normal images using low mean squared error (MSE). When defective regions occur at test time, high reconstruction error will be recorded since the model tries to reconstruct the defective regions only with normal patterns. Bergmann et al. [3] showed that loss of structural similarity (SSIM) can enhance localization to pixel-wise MSE in cutting across perceptual difference.

D. Patch Distribution Modeling (PaDiM).

Defard et al. proposed PaDiM [2], which is an important step in unsupervised anomaly detection. Notually learning to reassemble images, PaDiM predicts the distribution of patch-level features of a pretrained convolutional backbone. A multivariate Gaussian distribution is trained on the training

set features on each spatial location of the feature map. The Mahalanobis distance - which is a statistically principled measure computing anomaly scores observing feature correlations - is used to compute anomaly scores. PaDiM is always on the state of the art on MVTec AD, and ROC AUC values are more than 0.95 on a range of categories.

E. MVTec AD Dataset

Unsupervised anomaly detection With the introduction of the MVTec AD dataset [1] by Bergmann et al. at CVPR 2019, the de-facto benchmark in the field has become. It consists of 5,354 high-resolution photographs divided into 15 product categories in the industry (textured materials (carpet, leather, tile), structural object (bottle, capsule, pill), electronic component (cable, transistor). There are defect-free training images and test images in each category of 9 different training defect types annotated with pixel-precise ground truth masks.

III. METHODOLOGY

Each of the three works under the unsupervised environment: training is performed with defect-free images of all the categories and test-images having both normal and defective samples are evaluated. Deep learning methods have ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) used to normalize images to 256x256 pixels.

A. Data Pipeline

The implementation of a custom MVTecDataset class was used to index all image paths and ground truth mask paths in all 15 categories. The class can be used to load train and test splits, and provides the return of numpy arrays using classical algorithms and PyTorch tensors using a MVTecTorchDataset wrapper using deep learning algorithms. Ground truth masks Ground truth masks are binary images loaded with defective test images to perform localization assessment.

B. Classical Baseline: LBP + GLCM + OC-SVM

Each image is extracted to give four complementary features that make an 81-dimensional description. A 26 dimensional histogram encoding local micro-texture is obtained using the uniform rotation-invariant approach with radius=3 and npoints=24 in Local Binary Patterns (LBP). GLCM distances and angles[1,3] and angles=[0°, 45°, 90°] are used in 30 dimensions to represent five properties of contrast, dissimilarity, homogeneity, energy, and correlation. Sobel gradient statistics and canny edge density give 7 dimensions of structural edge information. Mean, standard, 25 th/75 th percentile, minimum, maximum of RGB values on a per-channel basis provide 18 dimensions.

StandardScaler is used to standardize the feature vectors, which are then reduced to 50 principal components by PCA - 99.9 % of the variance on average is retained. One-Class SVM (nu=0.1, gamma='scale') is performed with PCA features that are resulting. The negated score of SVM decision function is the anomaly score, where larger values imply more anomalousness.

C. Convolutional Autoencoder

The encoder of the convolutional model has some qualities with the classical autoencoders a 562,467 trainable

parameter symmetric convolutional autoencoder was written in PyTorch. The encoder has four blocks, with Conv2d (3x3 kernel, padding=1), BatchNorm2d, ReLU and MaxPool2d(2x2) blocks, in that order, successively reduced in spatial resolution 3x256x256 - 32x128x128 - 64x64x64 - 128x32x32 - 256x16x16. The decoder is based on this structure by applying ConvTranspose2d with stride=2 in order to upsample the decoder images, and finalizing with a Sigmoid activation, to keep the output in the range [0,1].

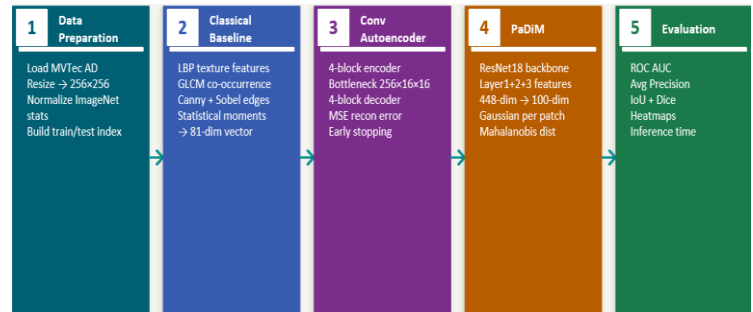
MSELoss (Adam, lr=1e-3, weightdecay=1e-5) and ReduceLROnPlateau scheduler (factor=0.5, patience=3) are used in training. Patience=7 early stopping is used, although the majority of categories were trained to completion (50 epochs). Each category is trained on a different model. The score of anomaly at the image level is the average per-pixel reconstruction error. Per- pixel error maps are localization heatmaps.

D. PaDiM: Patch Distribution Modeling

The frozen pretrained resnet18 backbone is used in PaDiM to extract multi scale patch features. The intermediates of the layer1 (64 channels, stride 4), layer2 (128 channels, stride 8) and layer3 (256 channels, stride 16) are extracted and upsampled to a shared spatial resolution of 64x64 using bilinear interpolation and complexed to 448 channel feature maps.

In order to cut down the computation cost, Sparse Random Projection [4] uses a projection of the 448-dimensional patch vectors on 100 dimensions to approximate the pairwise distance (Johnson-Lindenstrauss lemma). A Multivariate Gaussian $N(\mu_j^i, \Sigma_j^i)$ is then fitted at every spatial patch position of the 64x64 = 4,096 spatial positions with a covariance regularized with $\epsilon I = 0.01 \cdot I$ to provide numerical stability.

During inference, the Mahalanobis distance is calculated on the position of each patch (i, j) of a test image: $d(i,j) = \sqrt{((x-\mu_j^i)^T \Sigma_j^{i-1} (x-\mu_j^i))}$. The resultant 64x64 distance map is filtered using a Gaussian filter ($\sigma=4$) as well as upscaled to 256x256 using bilinear interpolation. The score of image-level anomaly is the largest distance between all patch positions.



IV. RESULTS

A. Experimental Setup

Each of the experiments was done in Google Colab with the NVIDIA A100-SXM4-80GB GPU (85.1 GB memory). Each of the 15 MVTec AD categories was independently run through the entire pipeline. ROC AUC and Average Precision (AP) are used to test the detection performance. Mean Intersection-over-Union (IoU) and Dice coefficient are compared to localize the anomaly. The mean

Intersection-over-Union (IoU) and Dice coefficient are calculated by thresholding the anomaly maps by the 90 th percentile.

B. Detection Performance

The scores of the three methods of ROC AUC in all 15 MVTec AD categories are shown in Table I. PaDiM has the greatest mean AUC of 0.9262 and is a winner in 12 / 15 categories. The classical OC-SVM has a mean AUC = 0.6468 - good on textured objects (tile: 0.919, toothbrush: 0.947) but dismal on even metallic surfaces (screw: 0.111, carpet: 0.291). The autoencoder performs poorly at an average AUC 0.5625 in the majority of the categorization, although it has a perfect AUC of 1.000 in screw the only category where it outperforms all the other methods.

TABLE I ROC AUC SCORES ACROSS ALL 15 CATEGORIES

Category	OC-SVM	AE	PaDiM
bottle	0.826	0.319	0.999
cable	0.654	0.595	0.863
capsule	0.648	0.545	0.914
carpet	0.291	0.875	0.961
grid	0.311	0.176	0.903
hazelnut	0.823	0.906	0.898
leather	0.722	0.675	0.999
metal_nut	0.617	0.468	0.978
pill	0.661	0.343	0.926
screw	0.111	1.000	0.690
tile	0.919	0.706	0.990
toothbrush	0.947	0.450	0.908
transistor	0.710	0.305	0.942
wood	0.654	0.735	0.991
zipper	0.810	0.341	0.930
Mean	0.647	0.563	0.926

AE = Convolutional Autoencoder. Bold values ≥ 0.95 . Mean row in gray.

C. Localization Performance

Table II is a report of the pixel-based localization measures of PaDiM. The average IoU of 0.2058 and average Dice of 0.3071 are obtained on all categories. There exists a strong correlation between performance and the size of the defect: bottle (IoU: 0.415) and tile (IoU: 0.411) have large and well defined defects, and screw (IoU: 0.031) has microscopic thread defects, less than 1 per cent of the image.

TABLE II PADIM LOCALIZATION METRICS (SELECTED CATEGORIES)

Category	Mean IoU	Dice
bottle	0.415	0.561
tile	0.411	0.557
cable	0.317	0.458

Category	Mean IoU	Dice
wood	0.264	0.391
metal_nut	0.254	0.387
zipper	0.240	0.370
hazelnut	0.208	0.319
carpet	0.203	0.302
transistor	0.185	0.295
toothbrush	0.173	0.264
pill	0.136	0.220
capsule	0.092	0.146
leather	0.086	0.143
grid	0.075	0.134
screw	0.031	0.059
Mean	0.206	0.307

Threshold at 90th percentile. Mean over all 15 categories shown.

D. Inference Time

A benchmark of inference time was done using 10 runs of each method on a single image. The autoencoder is the quickest with 1.4 ms/image (± 0.4 ms), thus, it is applicable in high throughput production lines. PaDiM has the best accuracy-speed trade-off of 10.3 ms/image (± 1.3 ms). Multi-scale feature extraction is the slowest at 88.2 ms/image (± 13.8 ms), which is the classical OC-SVM.

V. DISCUSSION

A. Why PaDiM Dominates

The advantages of PaDiM over its competitors are due to the quality of patch. ResNet18, which is trained on 1.2 million ImageNet images, represents deep hierarchical representations - edges and textures in the bottom, semantic patterns in the top. The combination of three scales (64+128+256 channels) offers the complementary information at various granularities. The multivariate Gaussian model is able to effectively model the joint distribution of co-occurring features at each spatial position but the Mahalanobis distance can take into consideration the feature associations and weigh each dimension within its variance.

B. Classical Method Limitations

The hand generated 81-dimensional feature vector is not good enough to differentiate between categories which have low texture difference between normal and defective areas. Thread defects on screw (AUC: 0.111) generate both LBP and GLCM values that are almost similar to normal thread patterns. On carpet (AUC: 0.291), color contamination defects transform global but local texture when LBP can be reliably used to detect it. This is one of the basic limits of handcrafted feature representations on complicated industrial surfaces.

C. Autoencoder Failure Analysis

The relatively low overall performance of the autoencoders (mean performance: 0.563) is in part attributable to the problem of reconstruction ambiguity: on complicated textured categories (carpet, grid), the autoencoder learns to recreate the overall pattern, with defective areas, which create low reconstruction error even on abnormal inputs. This is compounded on the issue of categories possessing a few training examples - toothbrush has just 60 training pictures. This is an exception with the ideal AUC on screw (1.000): the homogeneous metallic surface is not complicated enough that any thread scratch generates readily measurable reconstruction error.

D. Hyperparameter Sensitivity

In the case of PaDiM, the number of dimensions of the random projection (100) was selected so as to balance between computation and expressiveness. The smoothness $s=4$ of the Gaussian filter suppresses noise in anomaly maps, but has a tendency to blur small defects - one reason why the IoU of screw and grid are low. The localization point where the 90 th percentile falls is category-blind; category-based optimization would potentially boost scores of IoU significantly. In the case of the OC-SVM, $\nu=0.1$ was used as the anticipated fraction of outliers and was not category-dependent.

E. Success and Failure Cases

The most successful examples are bottle (PaDiM AUC: 0.999, IoU: 0.415) and leather (AUC: 0.999) where coarse texture training can result in close-to-perfect detection of large broken regions and contamination respectively. The classical SVM on screw (AUC: 0.111) where uniform texture properties are not able to discriminate the defect and the autoencoder on grid (AUC: 0.176) where the repeating pattern is rebuilt even with thread defects in it can be noted as examples of notable failures.

VI. CONCLUSION

In this paper, the comparison between the classical and the deep learning approaches to unsupervised industrial defect detection of the MVTec AD benchmark is presented in a systematic way. Our main results are: (1) PaDiM using pretrained deep features based on ResNet18, achieves mean ROC AUC of 0.9262, which is significantly higher than those of classical (0.6468) and autoencoders (0.5625) we have tried; (2) the transfer of pretrained deep features between ImageNet and industrial inspection is remarkably high without any task-specific training; (3) classical texture features hit a fundamental performance limit on uniform surfaces; (4) defect localization is still difficult with small defects.

Future avenues are: the per-category threshold optimization to enhance localization; the possibility to consider stronger backbones (WideResNet, EfficientNet); Gaussian fitting, as it can be done with the help of the GPUs, should be considered to reduce the training time of PaDiM; and finally, it is possible to refer to student-teacher network architecture. The essence of the knowledge, which is that transfer learning of general visual features allows highly effective unsupervised anomaly detection without defect samples, has wide general implication to quality control systems in actual manufacturing settings.

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REFERENCES

- [1] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD — A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 9592–9600.
- [2] T. Defard, A. Setkov, A. Loesch, and R. Audigier, "PaDiM: A Patch Distribution Modeling Framework for Anomaly Detection and Localization," in Proc. Int. Conf. Pattern Recognit. (ICPR), 2021, pp. 475–489.
- [3] P. Bergmann, S. Löwe, M. Fauser, D. Sattlegger, and C. Steger, "Improving Unsupervised Defect Segmentation by Applying Structural Similarity to Autoencoders," in Proc. Int. Conf. Comput. Vis. Theory Appl. (VISAPP), 2019, pp. 372–380.
- [4] D. Achlioptas, "Database-friendly random projections: Johnson-Lindenstrauss with binary coins," *J. Comput. Syst. Sci.*, vol. 66, no. 4, pp. 671–687, 2003.
- [5] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.
- [6] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst. Man Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.
- [7] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, 2001.
- [8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 770–778.
- [9] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in Proc. Int. Conf. Learn. Represent. (ICLR), 2015.